

Overview in General

This file contains a general overview of the data in the graph including node labels and relationships types.

References

- [jqassistant](#)
- [Neo4j Python Driver](#)

Node Labels

Table 1a - Highest node count by label combination

Lists the 30 label combinations with the highest number of nodes. The labels with the lowest node count are listed in table 1b. The total list would sum up to the total number of labels (100%).

The whole table can be found in the CSV report `Node_label_combination_count`.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Git, Update, Change]	26416	48.600813
1	[Git, Commit]	6744	12.407779
2	[Git, Change, Create]	5847	10.757456
3	[Git, Change, Delete]	4190	7.708866
4	[File, Git]	3576	6.579214
5	[Git, Change, Rename]	1699	3.125862
6	[Author, Git, Person]	1000	1.839825
7	[Git, Tag]	906	1.666881
8	[Json, Key]	668	1.229003
9	[Json, Value, Scalar]	603	1.109414
10	[Committer, Git, Person]	356	0.654978
11	[NPM, Dependency]	338	0.621861
12	[Type, TS, Declared]	294	0.540909
13	[Type, TS, Primitive]	294	0.540909
14	[TS, ExternalDeclaration]	215	0.395562
15	[Type, TS, Literal]	138	0.253896
16	[Json, Value, Object]	133	0.244697
17	[Type, TS, Union]	121	0.222619
18	[Type, TS, ObjectMember]	111	0.204221
19	[NPM, Script]	91	0.167424
20	[TS, Property]	65	0.119589
21	[TS, Function]	47	0.086472
22	[TS, Parameter]	44	0.080952
23	[Type, Object, TS]	42	0.077273
24	[Type, TS, FunctionParameter]	40	0.073593
25	[File, Directory]	34	0.062554
26	[Type, TS, Function]	34	0.062554
27	[Package, File, Json, NPM]	29	0.053355
28	[TS, Variable]	24	0.044156
29	[Value, TS, Literal]	20	0.036796

Chart 1a - Highest node count by label combination

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in Chart 1b.

<Figure size 640x480 with 0 Axes>

Nodes per label combination (more than 0.5% overall)

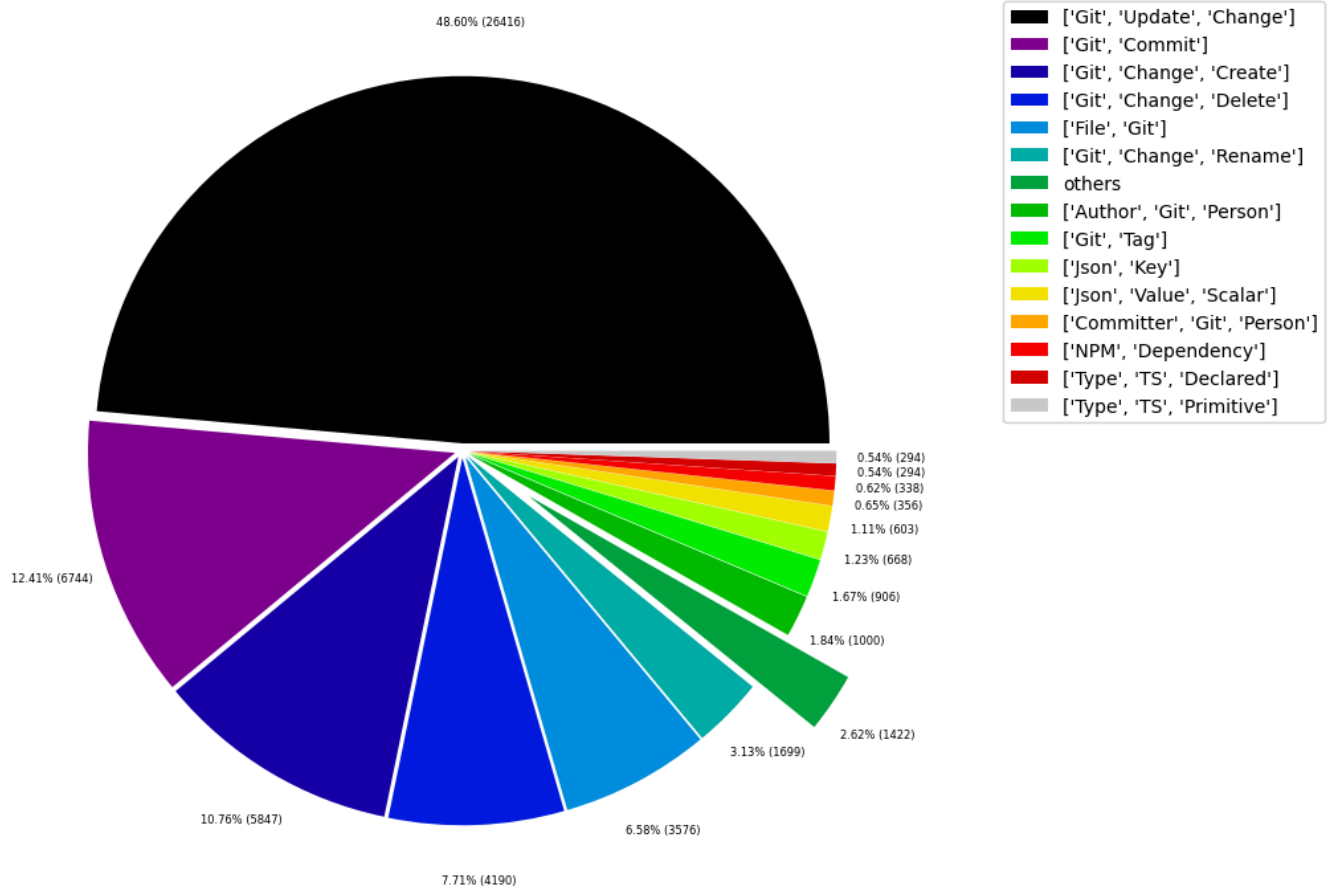


Table 1b - Lowest node count by label combination

Lists the 30 label combinations with the lowest number of nodes until they reach 0.5% of the total node count, which are shown above.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Analyze, Task, JQAssistant]	1	0.001840
1	[NPM, PublishConfig]	1	0.001840
2	[Value, TS, ObjectMember]	1	0.001840
3	[TS, Class]	1	0.001840
4	[Git, Branch]	1	0.001840
5	[TS, Method]	1	0.001840
6	[Repository, File, Git]	1	0.001840
7	[TS, Constructor]	1	0.001840
8	[TS, Enum]	2	0.003680
9	[Value, Object, TS]	3	0.005519
10	[Type, TS, Tuple]	3	0.005519
11	[Value, TS, Function]	4	0.007359
12	[TS, TypeParameter]	4	0.007359
13	[Repository, NPM]	5	0.009199
14	[File, TS, Scan]	5	0.009199
15	[File, Local]	5	0.009199
16	[Value, TS, Complex]	5	0.009199
17	[Project, TS]	5	0.009199
18	[Value, TS, Member]	6	0.011039
19	[File, TS, Local, Module]	6	0.011039
20	[Value, TS, Call]	6	0.011039
21	[Type, TS, TypeParameterReference]	6	0.011039
22	[NPM, Engine]	6	0.011039
23	[TS, EnumMember]	8	0.014719
24	[TS, ExternalModule]	11	0.020238
25	[Type, TS, NotIdentified]	11	0.020238
26	[Json, Value, Array]	12	0.022078
27	[Value, TS, Declared]	13	0.023918
28	[Git, Change, Copy]	15	0.027597
29	[TS, TypeAlias]	16	0.029437

Chart 1b - Lowest node count by label combination

Shows the lowest (less than 0.5% overall) node count label combinations. Therefore, this plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Nodes per label combination (less than 0.5% overall)

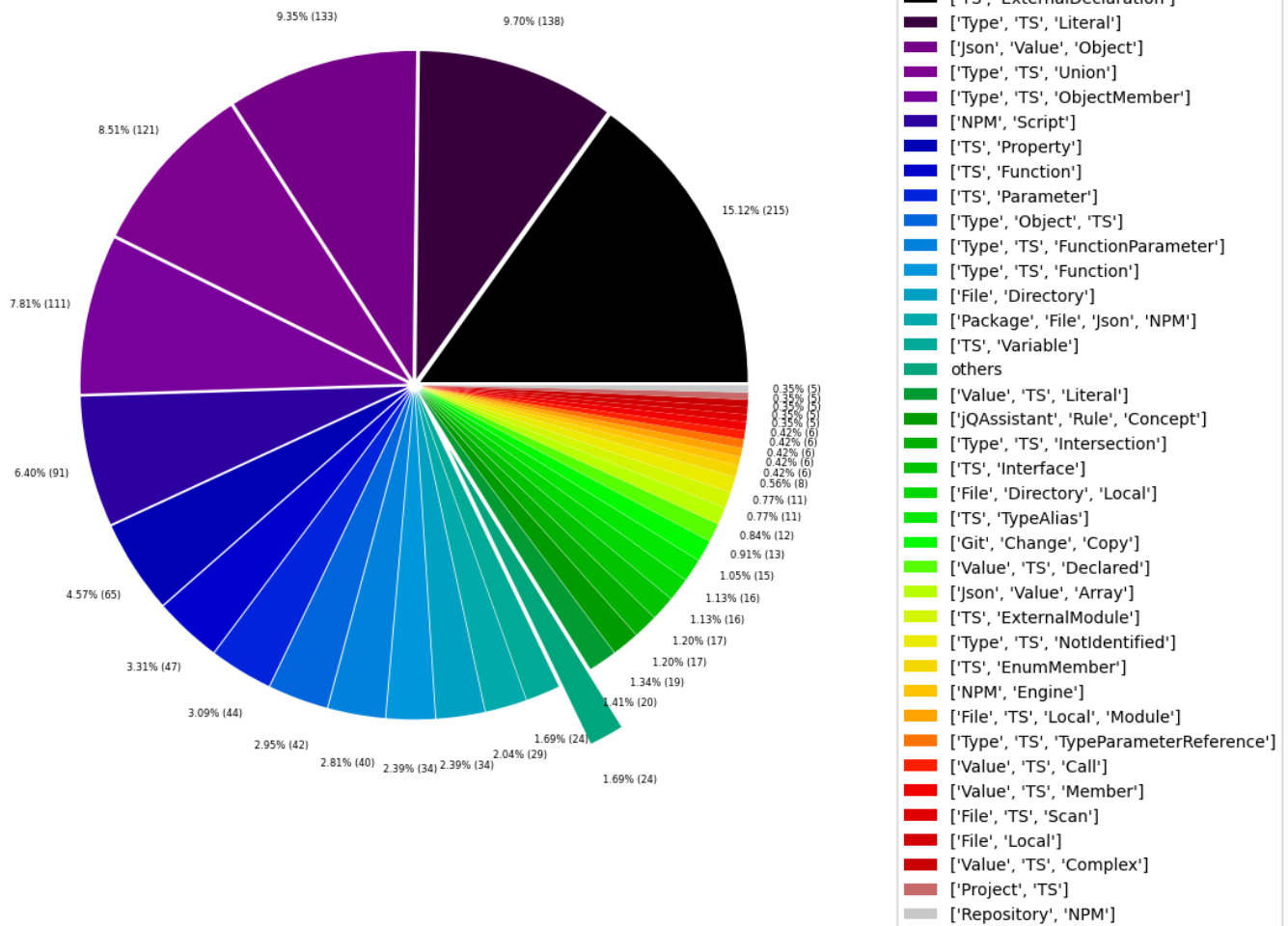


Table 1c - Highest node count by single label

Lists the 40 labels with the highest number of nodes. Doesn't sum up to the total number of nodes or 100% because one node can have multiple labels. Helps to identify commonly used labels.

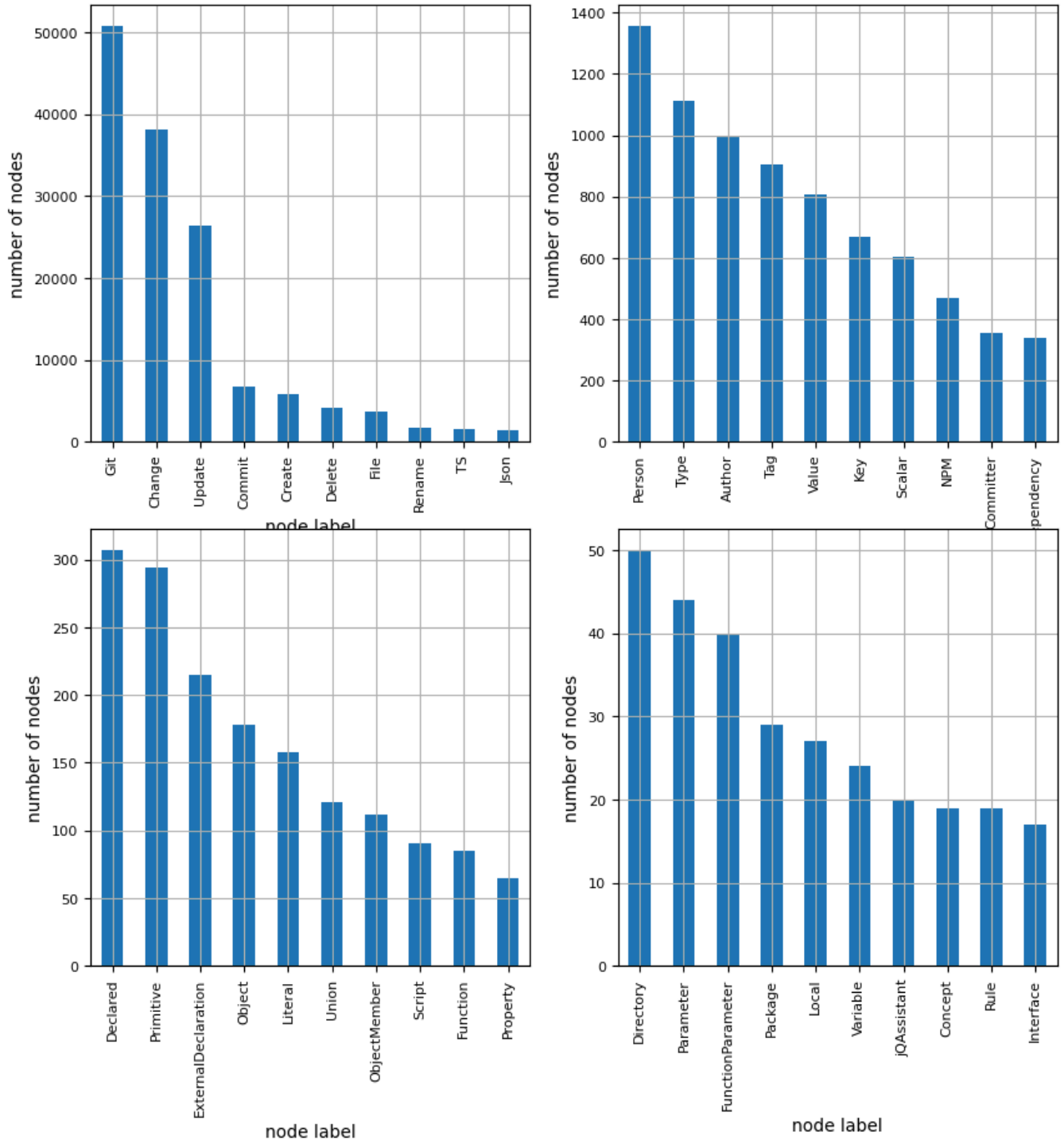
	nodeLabel	nodesWithThatLabel	nodesWithThatLabelPercent
0	Git	50751	93.372951
1	Change	38167	70.220595
2	Update	26416	48.600813
3	Commit	6744	12.407779
4	Create	5847	10.757456
5	Delete	4190	7.708866
6	File	3672	6.755837
7	Rename	1699	3.125862
8	TS	1641	3.019153
9	Json	1445	2.658547
10	Person	1356	2.494802
11	Type	1111	2.044045
12	Author	1000	1.839825
13	Tag	906	1.666881
14	Value	806	1.482899
15	Key	668	1.229003
16	Scalar	603	1.109414
17	NPM	470	0.864718
18	Committer	356	0.654978
19	Dependency	338	0.621861
20	Declared	307	0.564826
21	Primitive	294	0.540909
22	ExternalDeclaration	215	0.395562
23	Object	178	0.327489
24	Literal	158	0.290692
25	Union	121	0.222619
26	ObjectMember	112	0.206060
27	Script	91	0.167424
28	Function	85	0.156385
29	Property	65	0.119589
30	Directory	50	0.091991
31	Parameter	44	0.080952
32	FunctionParameter	40	0.073593
33	Package	29	0.053355
34	Local	27	0.049675
35	Variable	24	0.044156
36	jQAssistant	20	0.036796
37	Concept	19	0.034957
38	Rule	19	0.034957
39	Interface	17	0.031277

Chart 1c - Highest node count by label

Shows the 40 labels with the highest number of nodes.

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Node count by label



Relationship Types

Table 2a - Highest relationship count by type

Lists the 30 relationship types with the highest number of occurrences. The whole table can be found in the CSV report `Relationship_type_count`.

Total number of relationships: 160125

	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	CONTAINS_CHANGE	38167	23.835753
1	MODIFIES	38167	23.835753
2	UPDATES	26416	16.497112
3	COMMITTED	13488	8.423419
4	CREATES	7561	4.721936
5	HAS_PARENT	7517	4.694457
6	HAS_COMMIT	6744	4.211710
7	DELETES	5889	3.677752
8	HAS_FILE	3576	2.233255
9	RENAMES	1699	1.061046
10	CHANGED_TOGETHER_WITH	1691	1.056050
11	HAS_NEW_NAME	1032	0.644496
12	HAS_AUTHOR	1000	0.624512
13	HAS_TAG	906	0.565808
14	ON_COMMIT	906	0.565808
15	DEPENDS_ON	898	0.560812
16	HAS_KEY	668	0.417174
17	HAS_VALUE	668	0.417174
18	CONTAINS	598	0.373458
19	OF_TYPE	358	0.223575
20	HAS_COMMITTER	356	0.222326
21	EXPORTS	309	0.192974
22	REFERENCES	212	0.132397
23	DECLARES	186	0.116159
24	DECLARES_DEV_DEPENDENCY	169	0.105543
25	DECLARES_DEPENDENCY	161	0.100546
26	HAS_MEMBER	112	0.069945
27	HAS_TYPE_ARGUMENT	96	0.059953
28	DECLARES_SCRIPT	91	0.056831
29	HAS_PARAMETER	84	0.052459

Chart 2a - Highest relationship count by type

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in the second chart.

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Relationship types (more than 0.5% overall)

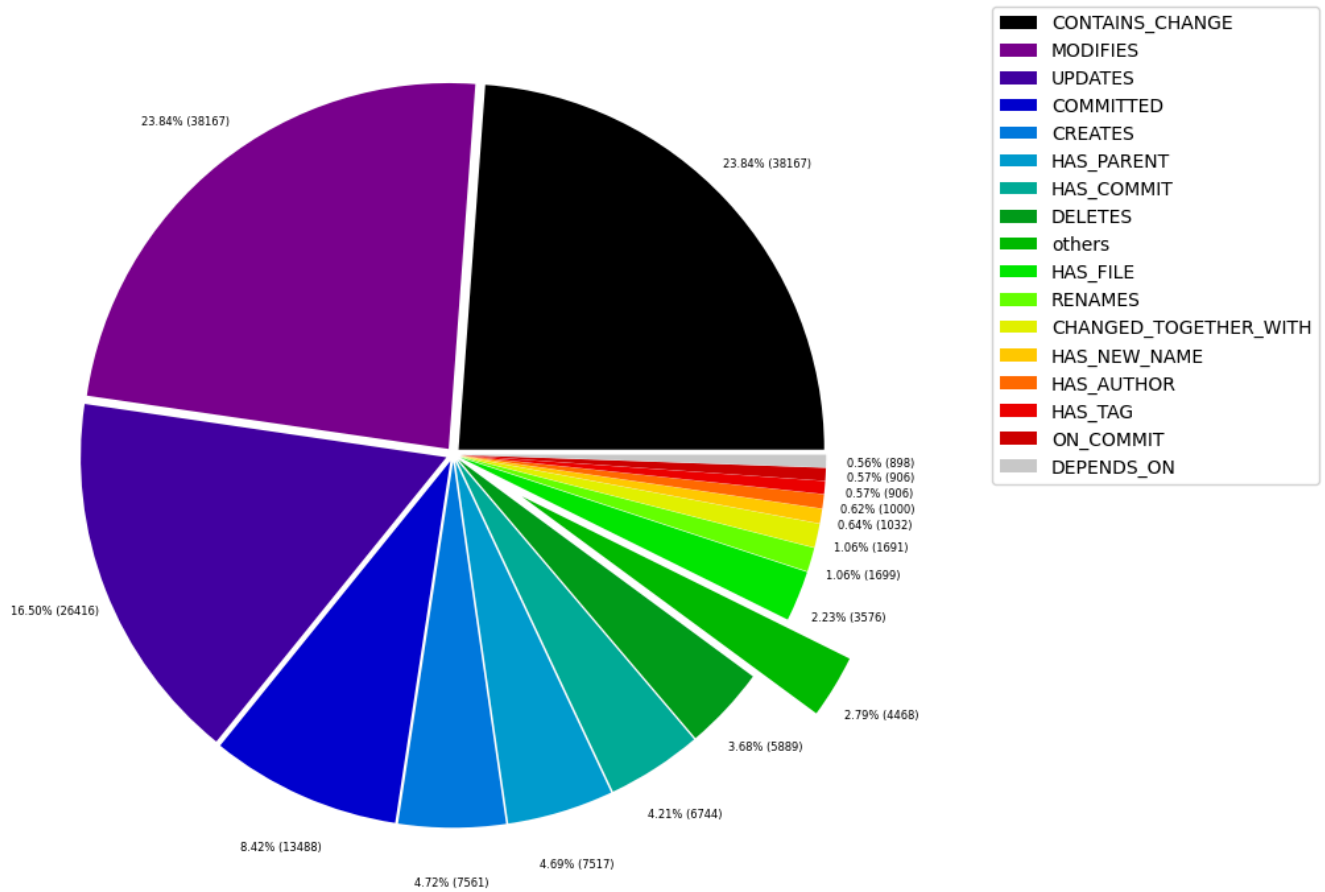


Table 2b - Lowest relationship count by type

Lists the 30 relationships type with the lowest number of occurrences up to 0.5% of the total node count. This is essentially breaking down the "others" slice from the chart above.

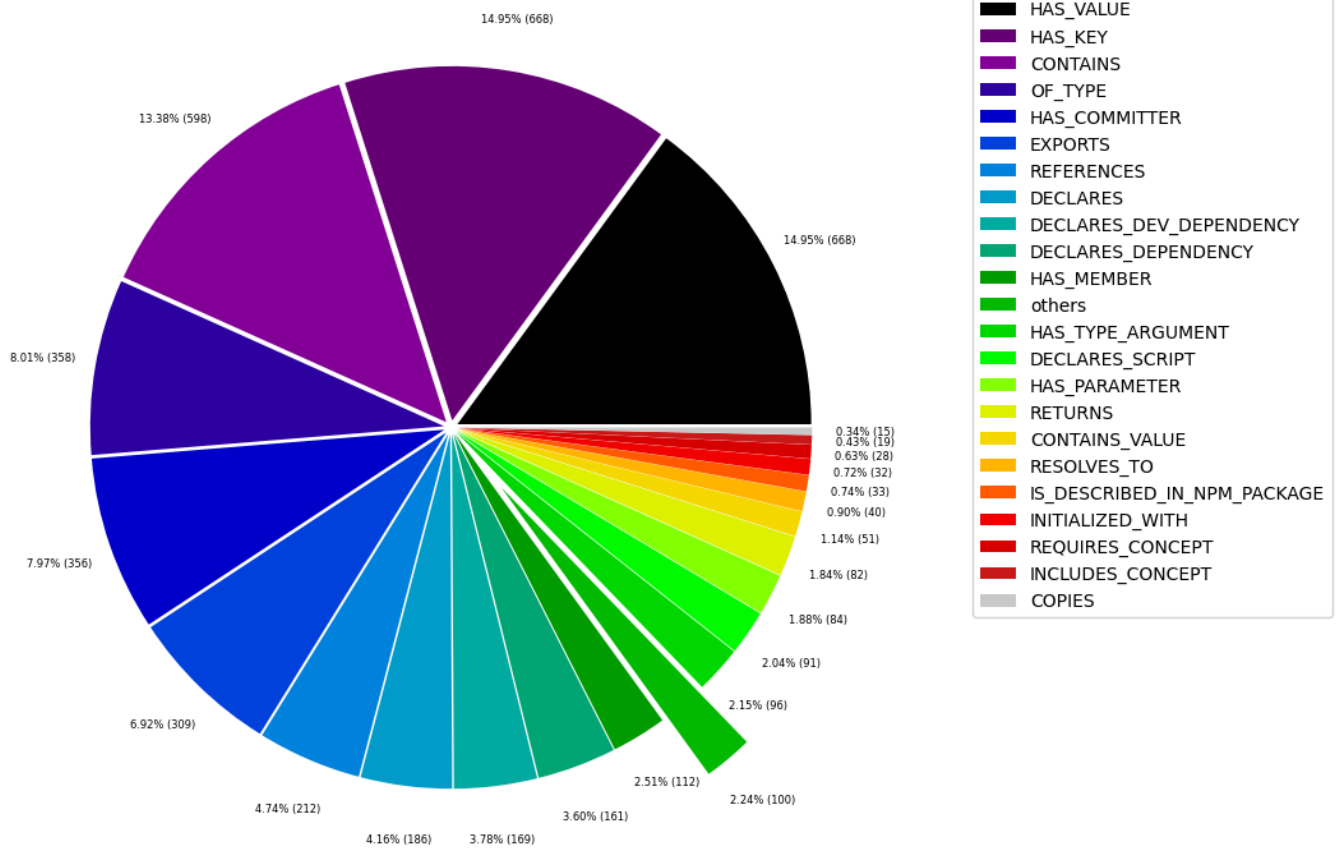
	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	HAS_BRANCH	1	0.000625
1	DECLARES_PUBLISH_CONFIG	1	0.000625
2	HAS_HEAD	2	0.001249
3	CONSTRAINED_BY	4	0.002498
4	IN_REPOSITORY	5	0.003123
5	HAS_ROOT	5	0.003123
6	HAS_NPM_PACKAGE	5	0.003123
7	HAS_CONFIG	5	0.003123
8	CONTAINS_PROJECT	5	0.003123
9	DECLARES_ENGINE	6	0.003747
10	EXTENDS	6	0.003747
11	CALLS	6	0.003747
12	MEMBER	6	0.003747
13	PARENT	6	0.003747
14	HAS_ARGUMENT	6	0.003747
15	DECLARES_PEER_DEPENDENCY	8	0.004996
16	USES	11	0.006870
17	COPY_OF	12	0.007494
18	COPIES	15	0.009368
19	INCLUDES_CONCEPT	19	0.011866
20	REQUIRES_CONCEPT	28	0.017486
21	INITIALIZED_WITH	32	0.019984
22	IS_DESCRIBED_IN_NPM_PACKAGE	33	0.020609
23	RESOLVES_TO	40	0.024980
24	CONTAINS_VALUE	51	0.031850
25	RETURNS	82	0.051210
26	HAS_PARAMETER	84	0.052459
27	DECLARES_SCRIPT	91	0.056831
28	HAS_TYPE_ARGUMENT	96	0.059953
29	HAS_MEMBER	112	0.069945

Chart 2b - Lowest relationship count by type

Shows the lowest (less than 0.5% overall) relationship types. This plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Relationship types (less than 0.5% overall)



Node labels with their relationships

Table 3a - Highest relationship count by node labels and relationship type

Lists the 30 node labels and their relationship types with the highest number of occurrences.

	sourceLabels	relationType	targetLabels	numberOfRelationships	numberOfNodesWithSameLabelsAsSource	numberOf
0	[Git, Commit]	CONTAINS_CHANGE	[Git, Update, Change]	26416	6744	
1	[Git, Update, Change]	MODIFIES	[File, Git]	26416	26416	
2	[Git, Update, Change]	UPDATES	[File, Git]	26416	26416	
3	[Git, Commit]	HAS_PARENT	[Git, Commit]	7517	6744	
4	[Repository, File, Git]	HAS_COMMIT	[Git, Commit]	6744	1	
5	[Author, Git, Person]	COMMITTED	[Git, Commit]	6744	1000	
6	[Committer, Git, Person]	COMMITTED	[Git, Commit]	6744	356	
7	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Create]	5847	6744	
8	[Git, Change, Create]	CREATES	[File, Git]	5847	5847	
9	[Git, Change, Create]	MODIFIES	[File, Git]	5847	5847	
10	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Delete]	4190	6744	
11	[Git, Change, Delete]	MODIFIES	[File, Git]	4190	4190	
12	[Git, Change, Delete]	DELETES	[File, Git]	4190	4190	
13	[Repository, File, Git]	HAS_FILE	[File, Git]	3576	1	
14	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Rename]	1699	6744	
15	[Git, Change, Rename]	DELETES	[File, Git]	1699	1699	
16	[Git, Change, Rename]	RENAMES	[File, Git]	1699	1699	
17	[Git, Change, Rename]	CREATES	[File, Git]	1699	1699	
18	[Git, Change, Rename]	MODIFIES	[File, Git]	1699	1699	
19	[File, Git]	CHANGED_TOGETHER_WITH	[File, Git]	1544	3576	
20	[File, Git]	HAS_NEW_NAME	[File, Git]	1032	3576	
21	[Repository, File, Git]	HAS_AUTHOR	[Author, Git, Person]	1000	1	
22	[Repository, File, Git]	HAS_TAG	[Git, Tag]	906	1	
23	[Git, Tag]	ON_COMMIT	[Git, Commit]	906	906	
24	[Json, Value, Object]	HAS_KEY	[Json, Key]	668	133	
25	[Json, Key]	HAS_VALUE	[Json, Value, Scalar]	552	668	
26	[Repository, File, Git]	HAS_COMMITTER	[Committer, Git, Person]	356	1	
27	[TS, Function]	DEPENDS_ON	[TS, ExternalDeclaration]	293	47	
28	[TS, ExternalModule]	EXPORTS	[TS, ExternalDeclaration]	215	11	
29	[Package, File, Json, NPM]	DECLARES_DEV_DEPENDENCY	[NPM, Dependency]	169	29	

Graph Density

total_number_of_nodes (vertices): 54353

total_number_of_relationships (edges): 160125

-> total directed graph density: 5.420259675706309e-05

-> total directed graph density in percent: 0.005420259675706309